

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	6
	3.3 SAR Based Analysis at Abutment 1 LHS	7
	3.4 SAR Based Analysis at Abutment 1 RHS	8
	3.9 SAR Based Analysis at Abutment 2 LHS	9
	3.10 SAR Based Analysis at Abutment 2 RHS	10
3.	Analysis	11
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 896+090

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

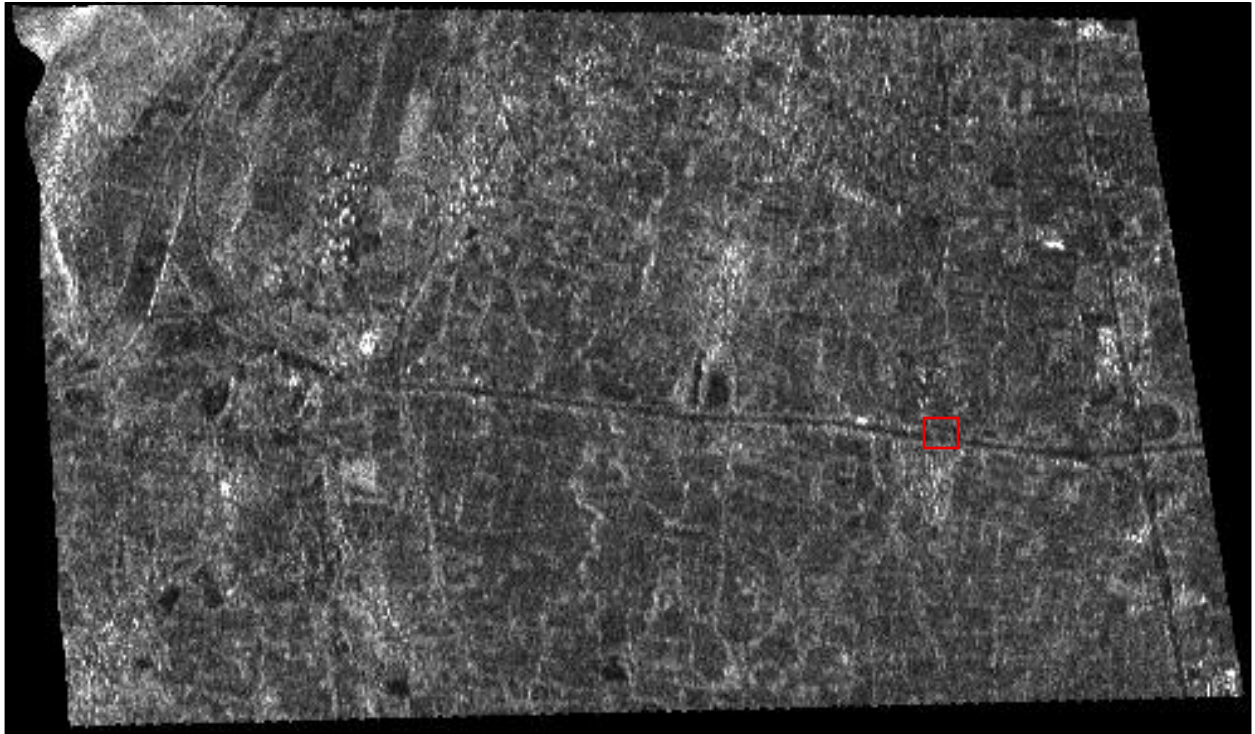
Flight Direction - DESCENDING

Polarization – VV+VH

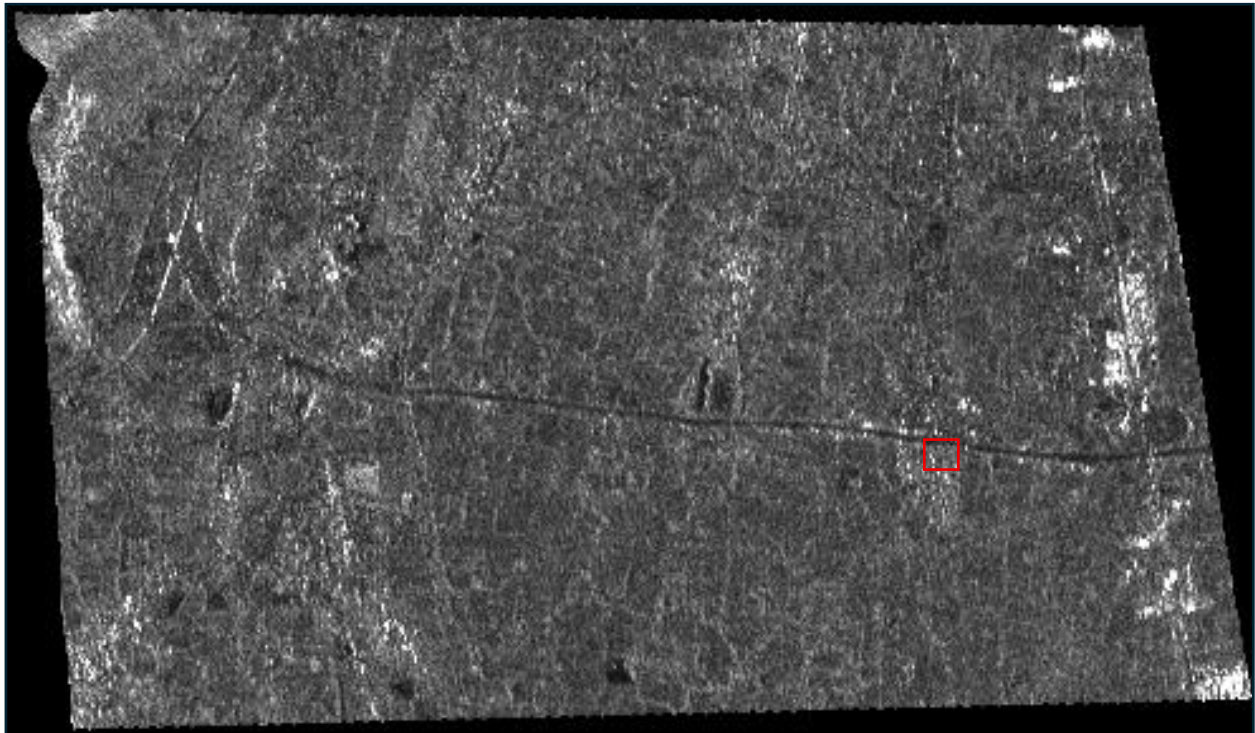
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

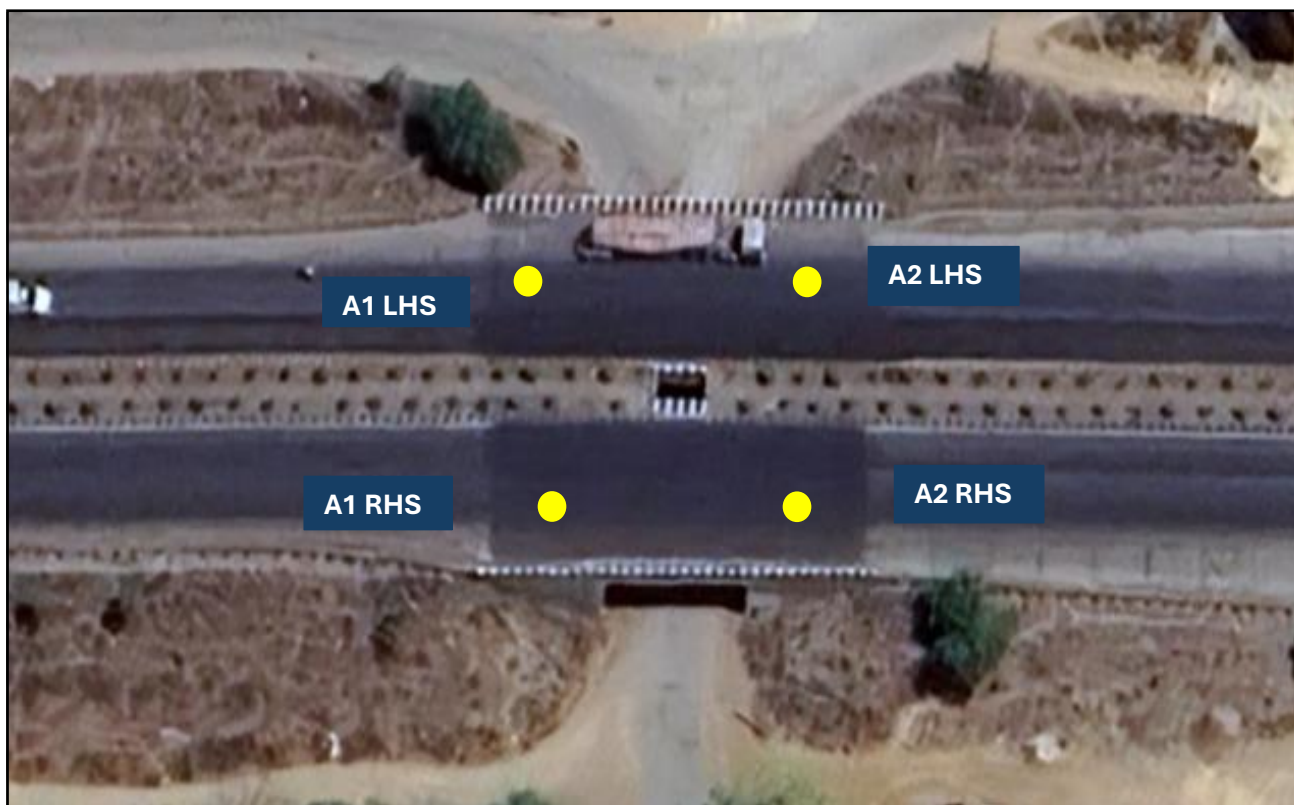


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

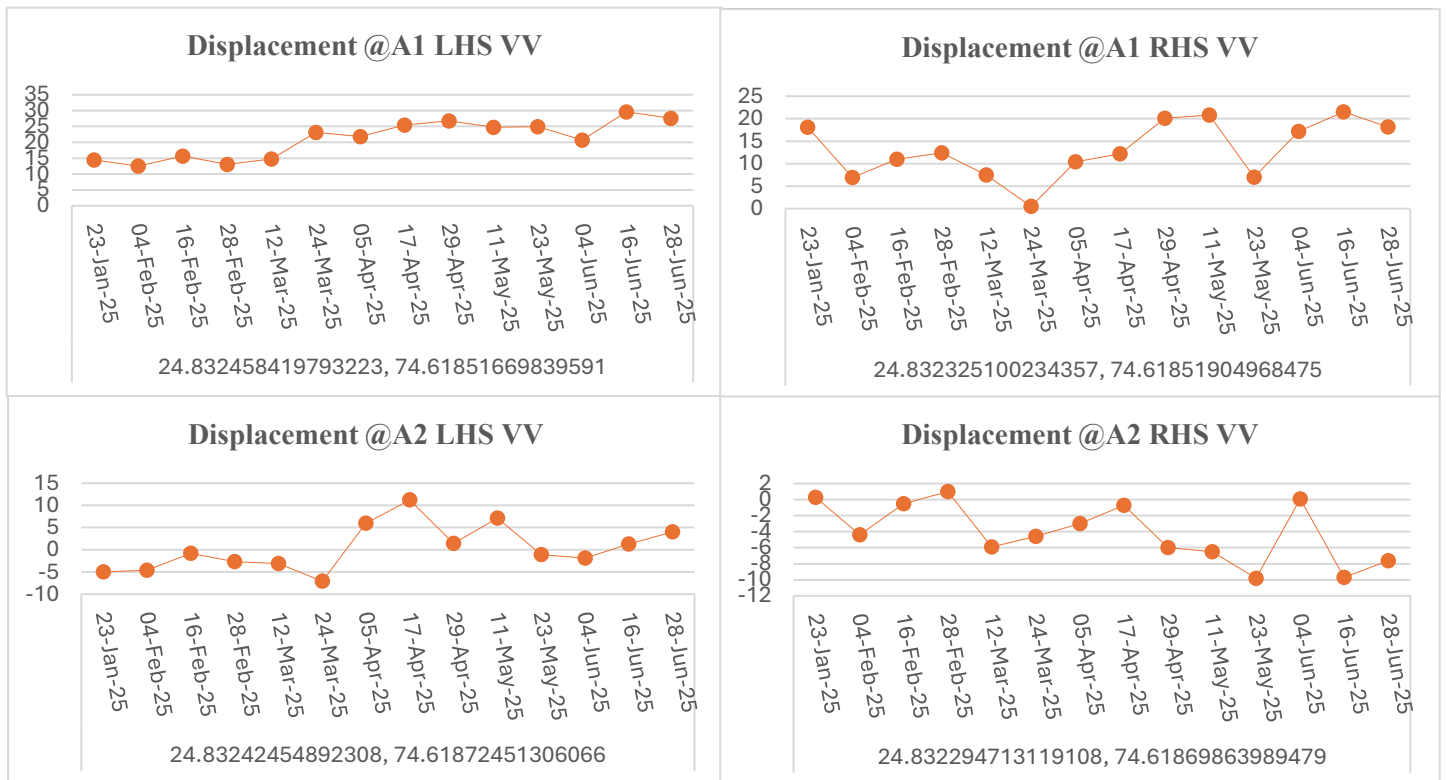
VUP – CH-896+090

Length – 13.77M

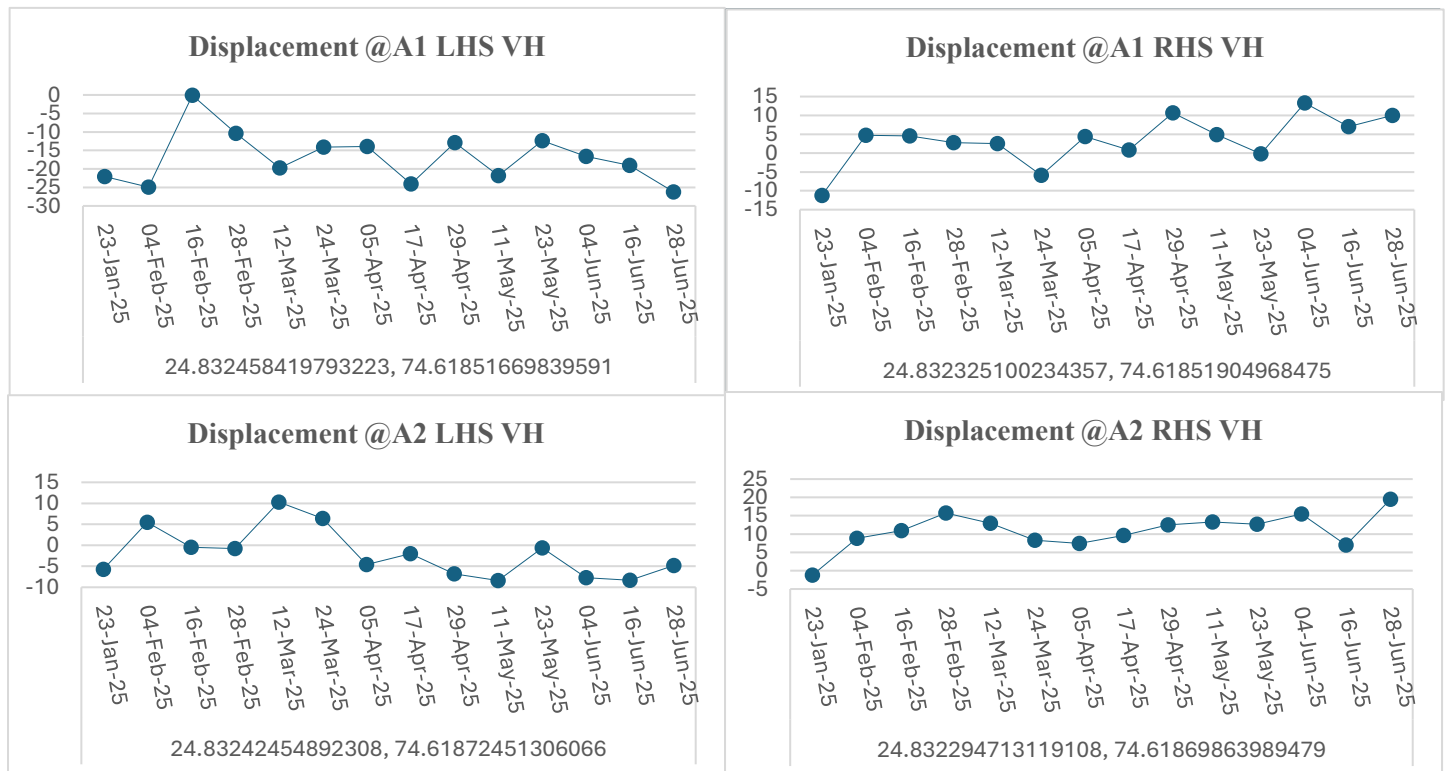


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



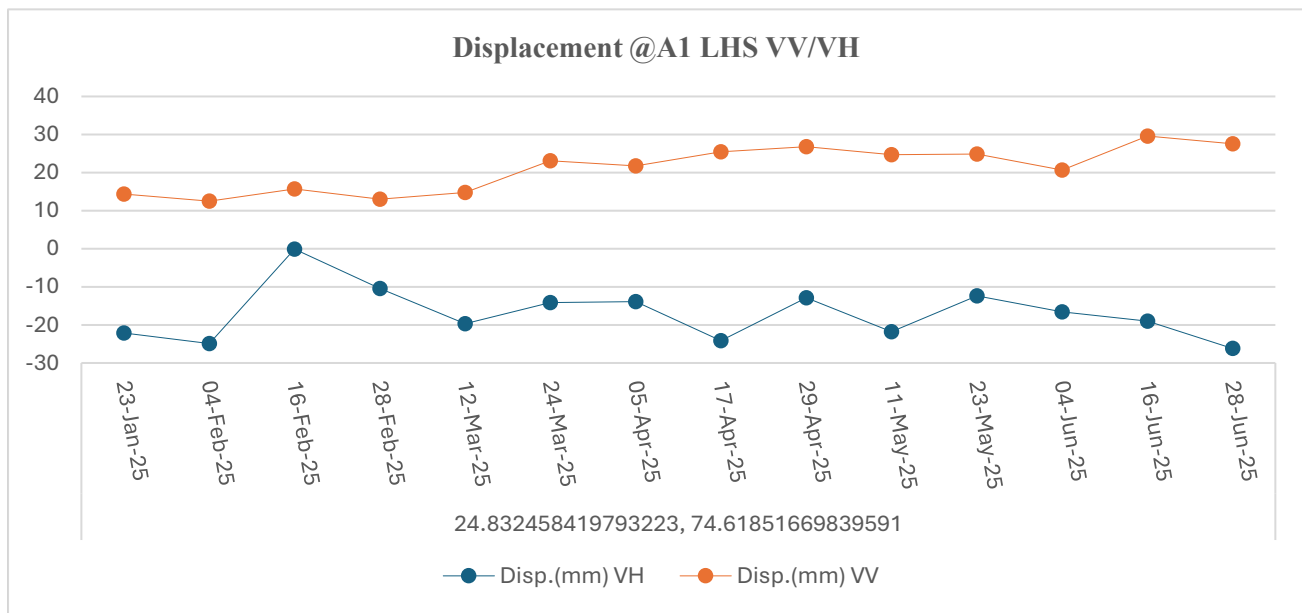
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.832458419793223, 74.61851669839591)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement is predominantly positive, varying approximately from +12 mm to +30 mm, indicating continuous uplift or upward movement with moderate variability over time.
- **Horizontal Displacement (VH)** Horizontal displacement at A1 LHS remains consistently negative throughout the monitoring period, ranging approximately from –5 mm to –30 mm.

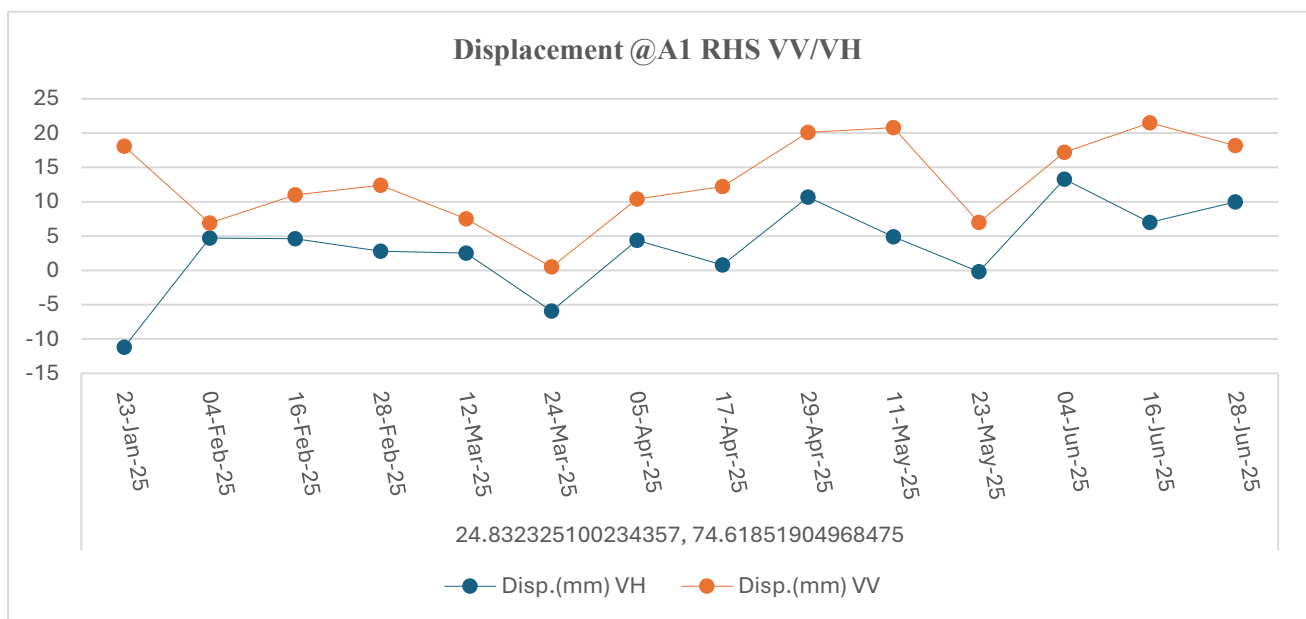
Critical Events & Anomalies:

- **February 2025:** VH shows a brief partial recovery toward –5 mm, while VV remains stable and positive, suggesting a temporary reduction in lateral displacement without affecting vertical uplift.
- **April 2025:** VH reaches one of its most negative values (–25 mm) concurrent with elevated VV (~+25 mm), marking a critical response phase with strong lateral displacement and sustained uplift.
- **June 2025:** VH drops further toward –30 mm, while VV peaks near +30 mm, indicating persistent lateral displacement accompanied by maximum vertical uplift, which may warrant closer long-term monitoring.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.832325100234357, 74.61851904968475)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement at A1 RHS exhibits a mixed trend, ranging approximately from -12 mm to +14 mm.
- **Vertical displacement (VV)** Vertical displacement remains predominantly positive, varying approximately from +1 mm to +22 mm, reflecting overall uplift with moderate temporal fluctuations.

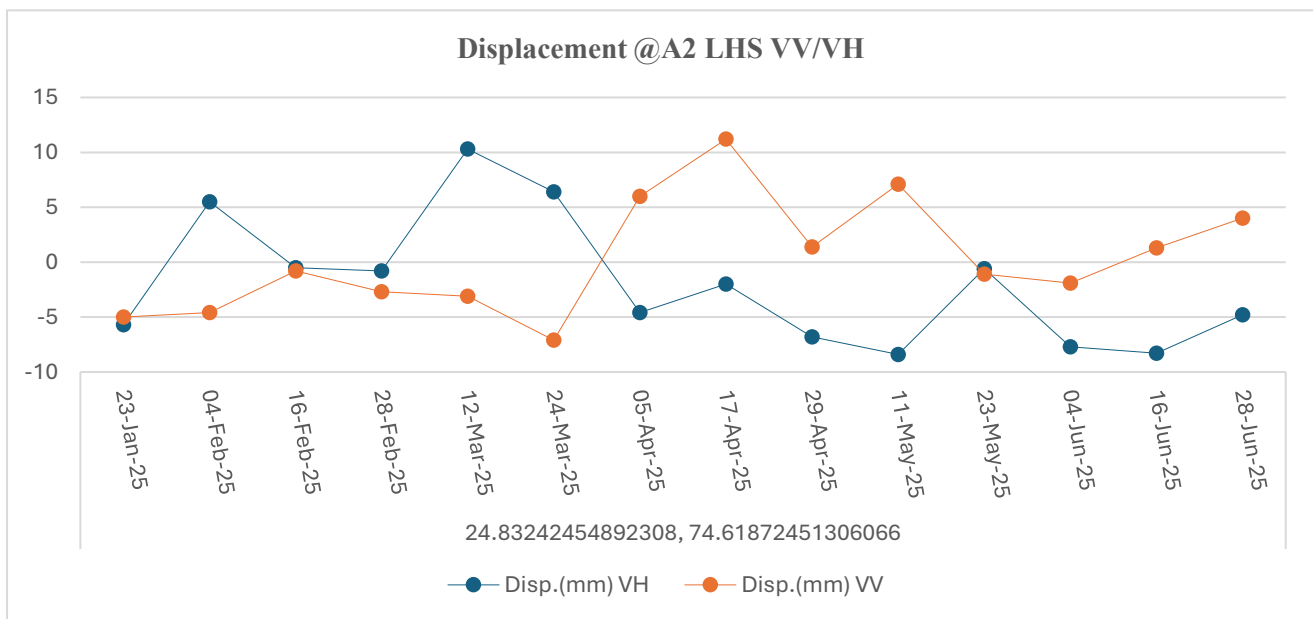
Critical Events & Anomalies:

- **March 2025:** VH reaches a local minimum (-6 mm) concurrent with the lowest VV (+1 mm), indicating a temporary combined reduction in both horizontal and vertical response.
- **May 2025:** VH shows renewed variability with a brief return toward negative values, whereas VV drops and then recovers, indicating short-term instability followed by re-stabilization.
- **June 2025:** VH stabilizes in the positive range (+8 to +14 mm) and VV remains consistently high (18–22 mm), suggesting lateral stabilization accompanied by sustained vertical uplift.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.83242454892308, 74.61872451306066)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement at A2 LHS shows a predominantly negative trend, ranging approximately from -9 mm to +10 mm.
- **Vertical Displacement (VV)** Vertical displacement varies approximately from -7 mm to +11 mm, reflecting moderate vertical movement with alternating settlement and uplift phases.

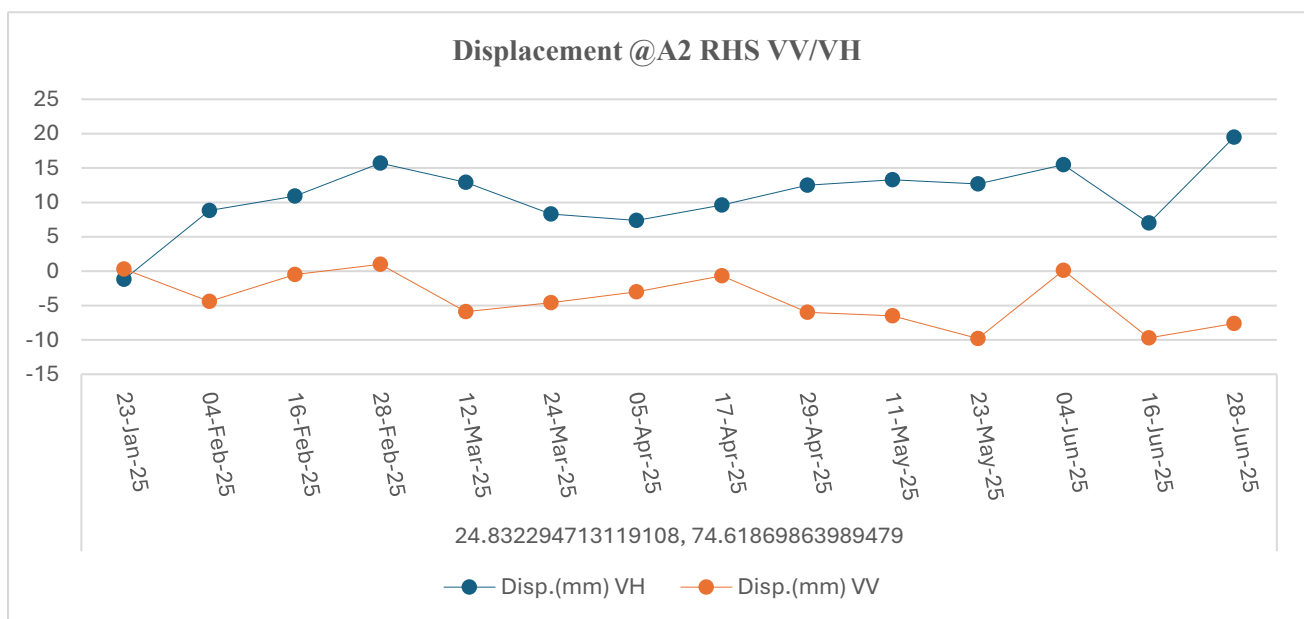
Critical Events & Anomalies:

- **February 2025:** VH reaches an early positive peak (+5 mm) while VV remains slightly negative, suggesting initial lateral response without corresponding uplift.
- **June 2025:** VH stabilizes in the negative range (-8 to -5 mm) while VV shows gradual recovery toward +4 mm, indicating lateral stabilization with mild vertical rebound

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.832294713119108, 74.61869863989479)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** VV is predominantly negative, varying roughly between -3 mm and -11 mm, suggesting ongoing settlement with intermittent partial recovery phases.
- **Horizontal (VH) displacement** VH remains consistently positive throughout the monitoring period, ranging approximately from +2 mm to +20 mm.

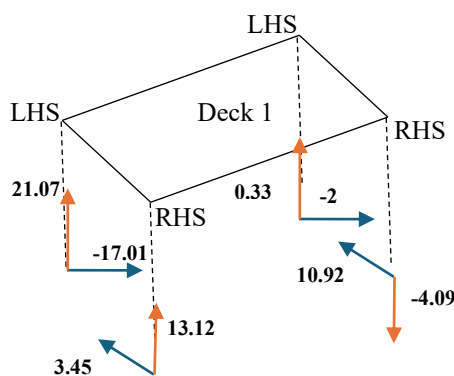
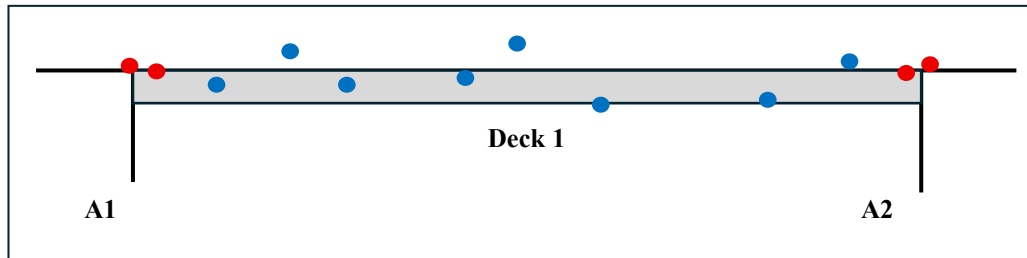
Critical Events & Anomalies:

- **February 2025:** VH shows a steady increase from low positive values (+3 mm) to an early peak of around +15 mm, while VV fluctuates slightly around zero to mildly negative values, indicating lateral movement preceding significant settlement.
- **June 2025:** VH shows notable variability dropping sharply to +7 mm mid-month before surging to its highest recorded value (+20 mm) by late June.

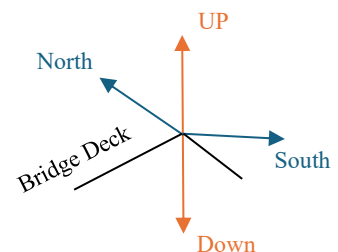
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab shows pronounced differential vertical displacement across LHS and RHS supports, confirming non-uniform deck response. The maximum uplift is observed at the LHS edge of Deck 1, reaching approximately +21.07 mm, representing the dominant vertical response in the system. This alternating uplift-settlement pattern highlights vertical distortion of the deck slab, suggesting uneven load transfer and support stiffness variation.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates clear directional variability, indicating asymmetric and torsional behaviour of the deck slab. The opposing directions of horizontal movement across LHS and RHS confirm torsional rotation of the deck rather than uniform lateral shift.

CONCLUSION:

- The bridge deck slab demonstrates pronounced asymmetric deformation, with peak vertical uplift at the LHS (+21.07 mm) and maximum horizontal displacement at the RHS mid-deck (+10.92 mm). The observed response underscores the importance of continued monitoring and detailed correlation with loading conditions, support stiffness, and construction stages to ensure long-term structural performance and serviceability.